6 Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score

data = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
   data.data, data.target, test_size=0.2, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)
pred = model.predict(X_test)

print("Confusion Matrix:\n", confusion_matrix(y_test, pred))
print("Accuracy:", accuracy_score(y_test, pred))
```

Output:

```
Naive Bayes Classification
Accuracy: 100.0 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

